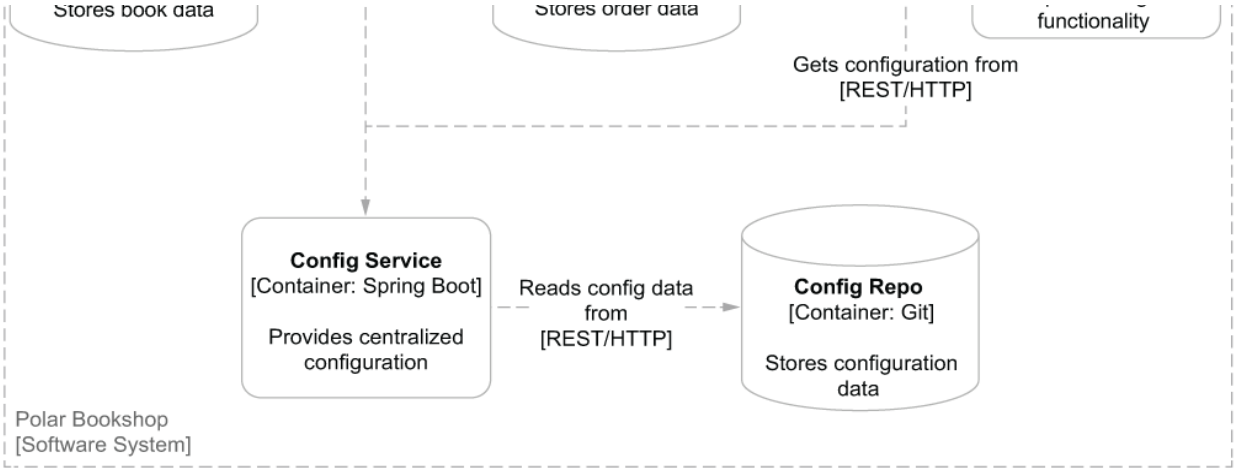


Figure 2.14 Architecture of Polar Bookshop. The cloud native system comprises application and data services with different responsibilities. For clarity, the security and observability services are not shown.

WEB AND INTERACTIONS

Polar Bookshop comprises several services that will have to communicate with each other to provide their functionality. You'll build RESTful services that interact synchronously over HTTP, both in a blocking way (using traditional servlets) and non-blocking ways (using reactive programming). Spring MVC and Spring WebFlux (on top of Project Reactor) will be your main tools for accomplishing such a result.

When building cloud native applications, you should design loosely coupled services and consider how you can keep data consistent in a distributed system context. Synchronous communications can generate problems when multiple services are involved in accomplishing a function. That's why event-driven programming is getting very popular in the cloud: it allows you to overcome the issues of synchronous communications.

I'll show you how to use events and messaging systems to decouple services and ensure data consistency. You'll use Spring Cloud Stream to implement